Notes of Modern Policy Learning Methods

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Outline

Intro
- Measure a policy space

Moving Towards PPO
- Intro
- (2015) Trust Region Policy Optimization (TRPO)
- (2015) Generalized Advantage Estimation (GAE)
- (2016) Advantage Actor-Critic (A2C)
- (2017) Proximal Policy Optimization (PPO)

LLM fine-tuning
- Intro
- (2022) Reinforcement Learning with Human Feedback (RLHF)
- (2023) Direct Policy Optimization (DPO)
- (2024) Group Relative Policy Optimization (GRPO)
- (2024) On-Policy Distillation (OPD)

Beyond MDP
- (2021) Decision Transformer

Discussion

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Intro (working)

This note mainly focus on some modern Policy Learning methods, including

- Policy based method
  - TRPO
  - PPO
  - ...
- Decision Transformer (Without MDP assumption)

RL-based view.

KL divergence

Two views of optimizing a policy-like-function.
Optimizing a Function v.s. Optimizing a Policy in RL sense

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Intro: Moving Towards PPO

- REINFROCE
- Trust Region Policy Optimization (TRPO)
  - can be seemed as a smart-conservative-greedy algorithm by **bounding the difference between old and new policy**
  - theoretically good but hard to implement (solving a constrained optimization problem is always hard)
- Proximal Policy Optimization (PPO)
  - similar goal as TRPO, but use a pessimistic bound, **clip on the goal directly instead of using KL-divergence constraint**
  - compared to TRPO, a little bit ad hoc, but more technically efficient, easy to implement
- Some algorithm refinement
  - GAE
  - A2C

Preliminaries: REINFORCE

(Williams 1992)(Sutton et al. 1999)

Recall Policy Gradient Theorem and REINFORCE with baseline,

$$\nabla_\theta V^{\pi_\theta}(s_1) = \mathbb{E}\left[Q^{\pi_\theta}(s,a)\nabla_\theta \log(\pi_\theta(s,a))\right]$$

$$= \mathbb{E}\left[\underbrace{\left(Q^{\pi_\theta}(s,a) - V^{\pi_\theta}(s)\right)}_{A(s,a)} \nabla_\theta \log(\pi_\theta(s,a))\right]$$

where

$$\mathbb{E} = \mathbb{E}_{s\sim d^{\pi_\theta}(s_1), a\sim\pi_\theta(\cdot|s)}$$

Its sample estimate is

$$\hat{R}_t^{(i)} := \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}^{(i)}$$

$$\hat{\mathbf{g}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T} \gamma^{t-1} \underbrace{\left(\hat{R}_t^{(i)} - b_t(s_t^{(i)})\right)}_{\hat{A}_t} \nabla_\theta \log(\pi_\theta(s_t^{(i)}, a_t^{(i)}))$$

For RL basics, see `https://oftrongs.github.io/RL/RL_summary.pdf`

Trust Region Policy Optimization (TRPO)

Introduce KL as policy distance measure(Schulman, Levine, et al. 2015)

TRPO extends Policy Improvement Theorem by replacing α with a distance measure between π and $\tilde{\pi}$.

$$\alpha \leftarrow D_{TV}^{max}(\pi, \pi^{new})$$

This theorem quantifies quantifies the least change of value for any two policies $\pi, \tilde{\pi}$ with policy advantage and KL-divergence of the two policies,

$$\eta(\tilde{\pi}) - \eta(\pi) \geq A_{\pi}(\tilde{\pi}) - \frac{2\gamma\varepsilon}{(1-\gamma)^2} D_{KL}^{\max}(\tilde{\pi}, \pi)$$

So we can design an Approximate Policy Iteration algorithm that guarantees increasing value:

In each iteration k = 0,1,2,... Solve the constrained optimization problem

$$\pi_{k+1} = \arg\max_{\pi} \left[A_{\pi_k}(\pi) - \frac{2\gamma\varepsilon}{(1-\gamma)^2} D_{KL}^{\max}(\pi_k, \pi) \right]$$

Trust Region Policy Optimization (TRPO)

Introduce KL as policy distance measure(Schulman, Levine, et al. 2015)

[[?]]
Now consider parameterized policies π_θ, and uses a constraint on the KL divergence rather than a penalty[1].

In each iteration $k = 0, 1, 2...$

$$\theta_{k+1} = \begin{array}{rl} \arg\max_\theta & \mathbb{E}_{s\sim\pi_{\theta_k}}\left[\sum_a \pi_\theta(s,a) A^{\pi_{\theta_k}}(s,a)\right] \\ \text{s.t.} & \mathbb{E}_{s\sim\pi_{\theta_k}}\left[D_{\mathrm{KL}}(\pi_\theta(\cdot|s), \pi_{\theta_k}(\cdot|s))\right] \leq \delta \end{array}$$

In practice, use the sample-based estimation of $\mathbb{E}_{s\sim\pi_{\theta_k}}$ and $A^{\pi_{\theta_k}}$ and the sample-based estimation of average KL-divergence.

$$\theta_{k+1} = \begin{array}{rl} \arg\max_\theta & \hat{\mathbb{E}}_{s\sim\pi_{\theta_k}}\left[\sum_a \pi_\theta(s,a) \hat{A}^{\pi_{\theta_k}}(s,a)\right] \\ \text{s.t.} & \hat{\mathbb{E}}_{s\sim\pi_{\theta_k}}\left[D_{\mathrm{KL}}(\pi_\theta(\cdot|s), \pi_{\theta_k}(\cdot|s))\right] \leq \delta \end{array}$$

[1] In practice, if using the penalty coefficient $\frac{2\gamma\varepsilon}{(1-\gamma)^2}$, the step sizes would be very small

Generalized Advantage Estimation (GAE)

how to estimate advantage using sample? (Schulman, Moritz, et al. 2015)

Suppose we sample M trajectories, with T_i time steps in trajectory i.
The total number of samples is $N = \sum_{i=1}^{M} T_i$
We have sample trajectories: [xx]

The empirical average can be written as

$$\hat{\mathbb{E}}_t \left[\rho_t(\theta) \hat{A}_t \right] = \frac{1}{N} \sum_{i=1}^{M} \sum_{t=0}^{T_i - 1} \rho_{i,t}(\theta) \hat{A}_{i,t}$$

where

$$\rho_{i,t}(\theta) = \frac{\pi_\theta(a_{i,t} \mid s_{i,t})}{\pi_{\text{old}}(a_{i,t} \mid s_{i,t})}.$$

Advantage Actor-Critic

how to sample? (Mnih et al. 2016)

how to get trajectories and rewards? a parallel style

Recall $\nabla_\theta V^{\pi_\theta}(s_1) = \mathbb{E}\left[\underbrace{(Q^{\pi_\theta}(s,a) - V^{\pi_\theta}(s))}_{A(s,a)} \nabla_\theta \log(\pi_\theta(s,a))\right]$,

In actor-critic method, the policy $\pi(\cdot,\cdot)$ is the actor and the baseline $V^\pi(\cdot)$ is the critic.

The question is: How we estimate this gradient, or How we estimate this Advantage?

A2C estimates this value through getting the trajectories by multi-agents using same policy in copy-worlds. This can efficiently get large sample, and we get a more robust estimate in statistical sense.

> "possible with both value-based and policy-based methods, off-policy as well as on-policy methods, and in discrete as well as continuous domains." (Mnih et al. 2016)

Proximal Policy Optimization (PPO)

(Schulman, Wolski, et al. 2017)

Let $\rho(\theta) := \frac{\pi_\theta(s,a)}{\pi_{\theta_{old}}(s,a)}$, recall that TRPO maximize (under constraint)

$$\theta_{new} = \arg\max_\theta \hat{\mathbb{E}}_{s,a\sim\pi_{\theta_{old}}} \left[\frac{\pi_\theta(s,a)}{\pi_{\theta_{old}}(s,a)} \hat{A}^{\pi_{\theta_{old}}}(s,a) \right],$$

$$L^{CPI}(\theta) = \hat{\mathbb{E}} \left[\rho(\theta)\hat{A} \right]$$

But TRPO is hard to compute(a constrained optimization problem), so an ad-hoc but efficient way comes:

PPO instead solve the unconstrained problem by maximize

$$L^{CLIP}(\theta) = \hat{\mathbb{E}} \left[\min \left(\rho(\theta)\hat{A}, \text{clip}(\rho(\theta), 1-\epsilon, 1+\epsilon)\hat{A} \right) \right]$$

Why clip-then-min?

a pessimistic bound (Schulman, Wolski, et al. 2017)

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}\left[\min\left(\rho(\theta)\hat{A}, \text{clip}(\rho(\theta), 1-\epsilon, 1+\epsilon)\hat{A}\right)\right]$$

- a pessimistic bound in the sense that
  - clip(ignore) when the policy improve
    - if $\hat{A} > 0$ and $\rho > 1+\epsilon$, the final is $(1+\epsilon)\hat{A}$
    - if $\hat{A} < 0$ and $\rho < 1-\epsilon$, the final is $(1-\epsilon)\hat{A}$
  - unclip when policy is worse
    - if $\hat{A} > 0$ and $\rho < 1-\epsilon$, the final is $\rho\hat{A}$
    - if $\hat{A} < 0$ and $\rho > 1+\epsilon$, the final is $\rho\hat{A}$

A soft way of conservative-greedy. By maximizing a pessimistic lower bound on the surrogate objective, PPO avoids the large jump to an estimated-better but actually-worse policy.

Proximal Policy Optimization (PPO)

(Schulman, Wolski, et al. 2017)

Algorithm 1 PPO, Actor-Critic Style

for iteration=$1, 2, \ldots$ **do**
  for actor=$1, 2, \ldots, N$ **do**
    Run policy $\pi_{\theta_{\text{old}}}$ in environment for T timesteps
    Compute advantage estimates $\hat{A}_1, \ldots, \hat{A}_T$
  end for
  Optimize surrogate L wrt θ, with K epochs and minibatch size $M \leq NT$
  $\theta_{\text{old}} \leftarrow \theta$
end for

Number of actors N, rollout length T, optimization epochs K, minibatch size $M \leq NT$, discount factor γ, GAE parameter λ, clipping parameter ϵ, coefficients c_V, c_H.

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Intro - Policy learning for LLM

modeling the MDP world for language

LLM is essentially a policy (on a MDP?)

- state - context (prompt x + already tokens $y_{<t}$), action - next token

$$s_t = (x, y_{<t}),\ \ a_t \sim \pi(\cdot \mid s_t)$$

- transition - deterministic, just add your decided token

$$\mathbb{P}(s_{t+1} = (x, y_{<t+1}) \mid s_t = (x, y_{<t}), a_t = y_t) = 1, \quad y_{<t+1} = (y_{<t}, y_t)$$

- A trajectory?

$$\tau = (s_1, a_1, s_2, a_2, s_3, a_3, ..., s_{T-1}, a_{T-1}, s_T)$$

$$\tau = ((x), (y_1), (x, y_1), (y_2), (x, y_1, y_2), (y_3), ..., (x, y_1, ..., y_{T-1}))$$

How large is LLM's state and action space? (No way to do MCMC!)

If vocab size = 100000, max output length = 1024k, for a fix prompt, we have

- size of action space: $|\mathcal{A}|$ is 100000
- size of state space: $|\mathcal{S}|$ is 100000^{1024k}

what about $|\pi|$?

Prob of a Trajectory

Let $D^\pi(\tau \mid x)$ denote the probability of a trajectory τ of states on starting from prompt x and following policy π.

$$\tau = (s_1, a_1, s_2, a_2, s_3, a_3, \ldots, s_{T-1}, a_{T-1}, s_T)$$

$$\tau = ((x), (y_1), (x, y_1), (y_2), (x, y_1, y_2), (y_3), \ldots, (x, y_1, \ldots, y_{T-1}))$$

- By the Markov property and chain rule:

$$D^\pi(\tau \mid x) \;=\; \prod_{t=1}^{T-1} \pi(a_t \mid s_t)\, \mathbb{P}(s_{t+1} \mid s_t, a_t)$$

- Degeneration in the LLM (deterministic transition) setting

  Let $y = (y_1, \ldots, y_{T-1})$:

$$D^\pi(\tau \mid x) \;=\; \prod_{t=1}^{T-1} \pi(y_t \mid x, y_{<t}) \;=\; \pi(y \mid x)$$

  so we have

~~bandit $\leftrightarrow$ per-step~~

If you take a bandit environment view, it presents a random user prompt and expects a response to the prompt.

Rethink LLM in a probability view

and what about its states/action distribution?
What LLM policy do?
-¿ generating a world
$D^{\pi}(\tau \mid x) =$

Reinforcement Learning with Human Feedback (RLHF)

A starting point (Christiano et al. 2017)(Ouyang et al. 2022)

For MDP $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P})$, $(\mathcal{S}, \mathcal{A}, \mathcal{P})$ settled, leaving $\mathcal{R}$ undefined...

Solution of RLHF paper:

(1) Defining (Training/Estimating) a reward model
- collect comparison data (prompt+win/lose output pair (x, y_w, y_l))
- train an additional reward model $r_\phi(\cdot, \cdot)$

(2) Use the reward model to define our $\mathcal{R}$eward of World $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P})$

$$R(s_t, a_t) = f(r_\phi(s_t, a_t))$$

DO NOT mix this two 'reward' !

(3) Then apply RL! by PPO, etc ...

choice of $r_\phi(\cdot, \cdot)$ in (Ouyang et al. 2022)'s practice
- Insight: humans are bad at scoring, but good at comparing
- a 6B model from Bradley-Terry (BT)
- critic: cannot capture non-transitive preferences

Reinforcement Learning with Human Feedback (RLHF)

A starting point (Christiano et al. 2017)(Ouyang et al. 2022)

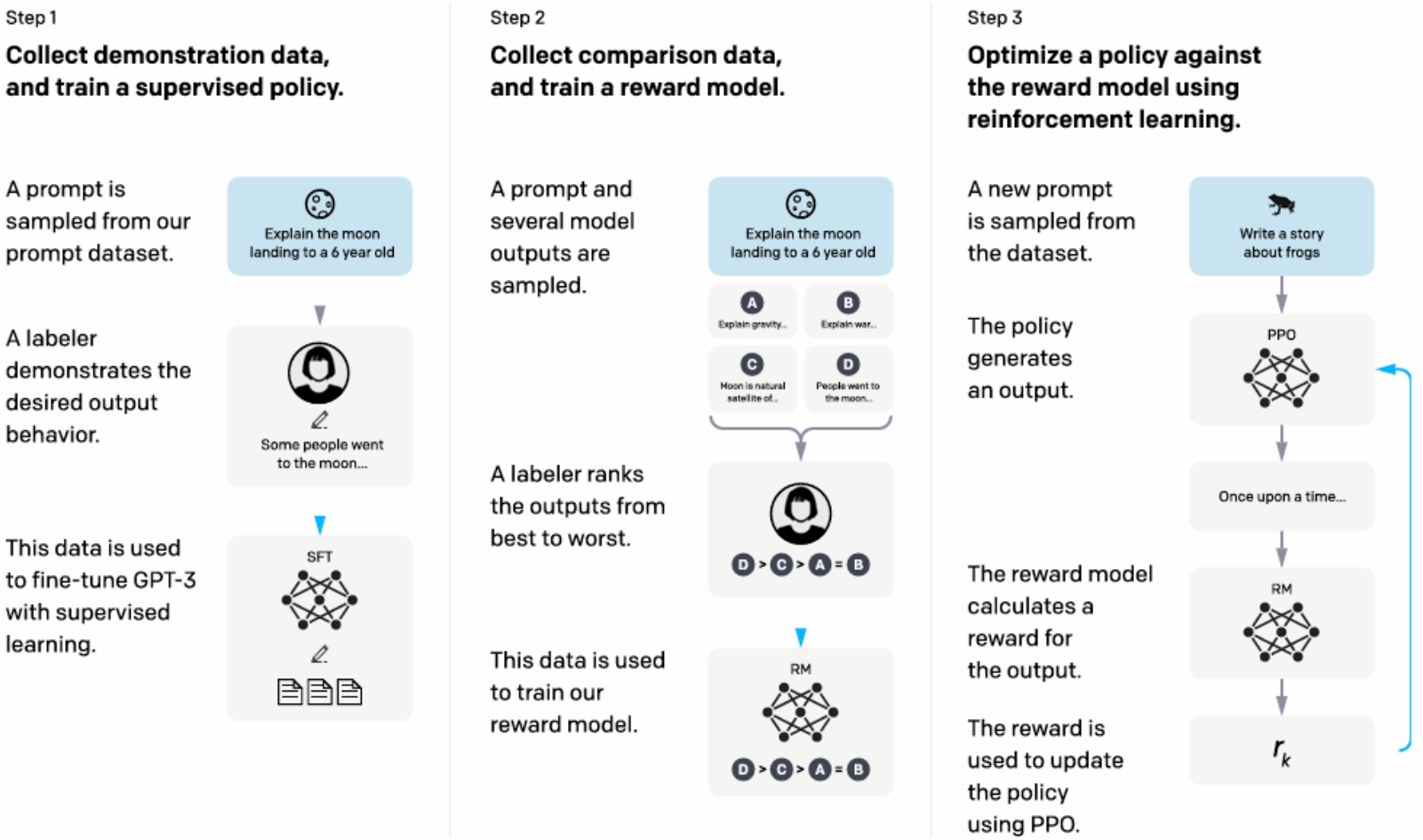


Figure 2: A diagram illustrating the three steps of our method: (1) supervised fine-tuning (SFT), (2) reward model (RM) training, and (3) reinforcement learning via proximal policy optimization (PPO) on this reward model. Blue arrows indicate that this data is used to train one of our models. In Step 2, boxes A-D are samples from our models that get ranked by labelers.

A special point of RL for LLM (diff from RL for RL)

constrained reward: Finetuning on a Base instead of Learning from Scratch

- per-token KL penalty into reward (dense reward version)

$$r_t = \begin{cases} -\beta \log \dfrac{\pi_\theta(y_t \mid x, y_{<t})}{\pi_{\text{ref}}(y_t \mid x, y_{<t})}, & t < T \\ r_\phi(x, y) - \beta \log \dfrac{\pi_\theta(y_t \mid x, y_{<t})}{\pi_{\text{ref}}(y_t \mid x, y_{<t})}, & t = T \end{cases}$$

From finite-horizon RL to the KL-regularized RLHF objective

Let $D^\pi(\cdot \mid x)$ denote the probability of a trajectory τ of states on starting from prompt x and following policy π.

$$\max_{\pi_\theta} \mathbb{E}_{x\sim P(Q),\ \tau\sim D^\pi(\cdot|x)}\left[\sum_{t=0}^{T} r_t\right] \quad \text{(classic finite horizon RL goal with } \gamma = 1)$$

$$= \max_{\pi_\theta} \mathbb{E}_{x\sim P(Q),\ \tau\sim D^\pi(\cdot|x)}\left[r_\phi(x, y) - \beta \sum_t \log \frac{\pi_\theta(y_t|s_t)}{\pi_{\text{ref}}(y_t|s_t)}\right]$$

$$= \max_{\pi_\theta} \mathbb{E}_{x\sim P(Q),\ \tau\sim D^\pi(\cdot|x)}\left[r_\phi(x, y)\right] - \beta\, \mathbb{D}_{\text{KL}}\left(\pi_\theta(y|x) \,\|\, \pi_{\text{ref}}(y|x)\right)$$

RLHF MDP with PPO style implementation

Direct Policy Optimization

Back to a SFT style finetuning (Rafailov et al. 2023)

Maintaining a reward function is also expensive, so 'skip' the reward function and optimize directly on offline data (x, y_w, y_l) pairs

DPO goal, pair-wise DPO loss:

$$\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right]$$

Pros and Cons of DPO

- easy to implement
- offline so can not get possibly-better trajectory samples (compared with online ways)

Group Relative Policy Optimization

byebye critic, a smart MC (Shao et al. 2024)

Use group-wise advantage estimate, i.e. the relative reward advantage of multiple sampled outputs, produced in response to the same question

$$\hat{A}_{i,t} = \tilde{r}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}$$

$$I_{i,t}(\theta) = \min\left[\rho_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}\left(\rho_{i,t}(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_{i,t}\right] - \beta\,\mathbb{D}_{\text{KL}}\left[\pi_\theta \| \pi_{\text{ref}}\right]$$

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x\sim P(Q),\ \{y_i\}_{i=1}^{G}\sim D^{\pi_{\theta_{\text{old}}}}(\cdot|x)}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|y_i|}\sum_{t=1}^{|y_i|} I_{i,t}(\theta)\right]$$

Why GRPO here?

- No separate value model, resource-efficient in compute and memory sense
- Characteristic of math tasks: a more final-answer-means-everything task

On-Policy Distillation

(R. Agarwal et al. 2024)

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Decision Transformer

a new framework for offline RL task

Decision Transformer(DT) reformulates the offline RL task as **a sequence modeling problem**.

- the return-to-go(sum of future rewards) $\hat{R}_t = \sum_{t'=t}^{T} r_{t'}$
  - reformulated trajectory: $(\hat{R}_1, s_1, a_1, ..., \hat{R}_T, s_T, a_T)$
- Goal: learn a prediction model that predicts the action based on past returns-to-go, states, and actions(analogous to a language modeling task! action = next word)

$$\hat{a}_t = f_\theta(\hat{R}_{\leq t}, s_{\leq t}, a_{<t})$$

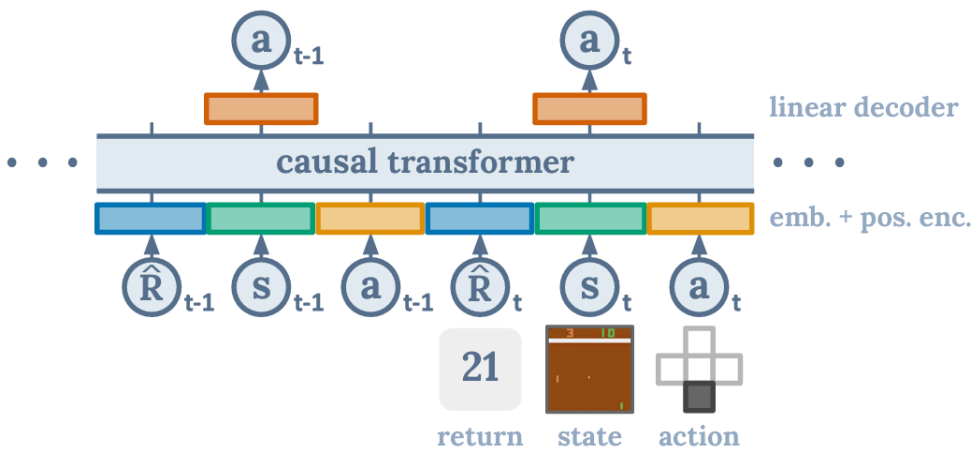


Figure: Decision Transformer architecture

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What does your goal like (do you need MDP assumption? is your action space continuous or discrete? finite horizon or infinite horizon or ...? do you need a deterministic policy or also good with stochastic one?)
What does your policy look like?

Optimizing a Function v.s. Optimizing a Policy

$\max f(x)$
$\max V^{\pi}(s)$

Online v.s. Offline, On-policy v.s. Off-policy

- Online RL v.s. Offline RL: can/can not interact with the environment to get new sample trajectory [?]

Estimating Advantage: MC or by Critic?

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